

ORB

Omni Reconnaissance Bot

YOLO-based plan for human + weapons (knife, gun, other handheld weapons) detection on a Raspberry Pi 4

High-level architecture (what you'll build)

- Two cameras → capture frames (synchronized-ish) → preprocessor → YOLO detector (person, gun, knife, other weapons) → postprocess (NMS, confidence threshold) → overlay + stream to Flask/WebRTC dashboard + alert system (MQTT / buzzer).

Software stack & libraries

- Raspberry Pi OS (64-bit) or Ubuntu 22.04 64-bit (recommended for ONNX/OpenVINO support).
- Python 3.9+ virtualenv
- OpenCV (with libcamera / v4l2 support)
- Ultralytics ultralytics (for training/export) — training on desktop/GPU, not Pi. [Ultralytics Docs](#)
- ONNX Runtime and/or OpenVINO for acceleration on Pi. ONNX Runtime has Pi guides; OpenVINO provides optimization flow for YOLOv8

Dual-camera inference program (Python)

- handles two camera streams (CSI via libcamera or v4l2 and USB via /dev/video*),
- runs ONNX Runtime inference (single-threaded or multithreaded),
- does preprocessing and postprocessing (NMS),
- draws boxes and sends frames to a Flask MJPEG endpoint.

Performance tuning & optimization tips

- Use yolov8n (nano) for edge. If speed allows, try yolov8s.
- Export to ONNX then run with OpenVINO or ONNXRuntime on Pi for better perf. OpenVINO has tutorials for optimizing YOLOv8. [OpenVINO Documentation+1](#)
- Quantize model (FP16 / INT8) if supported by conversion tool — reduces CPU ops.
- Use frame-skipping: run inference on every nth frame (e.g., process at 5–8 FPS, capture at 15–30 FPS).
- Use threading/process pools for capture + inference to keep cameras streaming while inference runs.
- Reduce input imgsz to 320–640 depending on tradeoff.

Raspberry Pi Setup

2.2 Boot Pi & Update

Insert the SD Card → Power ON → Open Terminal

```
sql

sudo apt update && sudo apt upgrade -y
```

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Enable cameras and interfaces:

```
arduino

sudo raspi-config
```

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Enable following:

- ✓ Camera (CSI)
- ✓ I2C (thermal sensor)
- ✓ SPI (optional)
- ✓ SSH
- ✓ VNC (optional)

Reboot:

```
nginx

sudo reboot
```

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